A Structural Conjecture for the Primary Gravitational Leakage Parameter in the PDL Framework: Towards a Parameter-Free Derivation of Newton's Constant

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Abstract

Within the Projective Dynamic Logo (PDL) framework, Newton's gravitational constant G is currently predicted via the bridge formula [4], which requires the primary coherence leakage parameter ε_G as an intermediate. Although ε_G is constrained structurally through its relation to the proton quintuplet, it is ultimately calibrated from the empirical value of G. The present paper reports a numerical conjecture that expresses ε_G directly and entirely in terms of structural quantities of the proton architecture: the surface engagement fraction $\kappa = R_{\text{surf}}/R_{\text{tot}}$, the no-self-loop coherence correction $\mathcal{C} = (n_u^2 - 1)/n_u^2 = 575/576$, and the total relational budget $R_{\text{tot}} = 11017$. The conjecture reads

$$\varepsilon_G \stackrel{?}{=} \frac{\kappa \cdot \mathcal{C}}{6+\kappa} \cdot \left(1 + \frac{2}{R_{\text{tot}}}\right),$$

and is exact on $\mathbb{Q}(\sqrt{5})$. Its numerical value agrees with the CODATA 2022 determination to **17 ppm**, well within the spread of independent experimental measurements of G, which currently spans approximately 500 ppm [17, 14, 15, 12, 3]. We document the convergence series leading to this conjecture, provide a structural interpretation of each factor, and reduce the proof to a single open problem: the derivation of the proton-to-electron mass ratio μ from the PDL axioms. We show that the conjecture is equivalent to $\mu_{\text{exp}} = \mu^*$, where $\mu^* = 1836.152670$ is a value in $\mathbb{Q}(\sqrt{5})$ that agrees with the CODATA 2022 value $\mu_{\text{exp}} = 1836.152673$ to 0.002 ppm — four orders of magnitude inside the current experimental precision of 0.000017 ppm. The 17 ppm gap between the conjecture and CODATA is identified as the signature of the dynamical dressing of the structural (bare) mass ratio $\mu_{\text{PDL}} = R_{\text{tot}}/6 = 11017/6$ by coherence leakage corrections. If proved, this conjecture would make ε_G, and hence G, a fully combinatorial prediction from the proton quintuplet $(24, 28, 930, 10087, 11017)$ without any gravitational empirical input.

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1 Introduction

The Projective Dynamic Logo (PDL) framework reconstructs physical reality from four axioms on finite signed graphs: minimal binary pulsation, triangular coherence, minimal completeness, and logical optimisation [10, 8]. The proton is a hierarchical composite characterised by the unique integer quintuplet $(n_u, n_d, r_{\mathrm{val}}, R_{\mathrm{sea}}, R_{\mathrm{tot}}) = (24, 28, 930, 10087, 11017)$ [5]. From this quintuplet, the fine-structure constant α [7] and Newton's gravitational constant G [4] are derived without free parameters, with residuals of 1.0×10^{-4} and 27 ppm respectively against CODATA 2022 [17]. The exponent 18 governing the gravitational hierarchy has been established as a topological invariant of $S^2 = \partial\Delta^3$ [11].

Despite these results, one open problem remains at the foundation of the gravitational sector: the primary coherence leakage parameter ε_G, defined by $Gm_p^2/(\hbar c) = \varepsilon_G^{18}$, is not yet derived from the proton quintuplet alone. In D24 [4], ε_G is extracted from the bridge formula

$$\Delta r_{\mathrm{val}} \;=\; \varepsilon_G \cdot \mathcal{R} \cdot \mathcal{C}, \qquad \mathcal{R} = \frac{R_{\mathrm{tot}}\, R_{\mathrm{surf}}}{r_{\mathrm{val}}^2}, \quad \mathcal{C} = \frac{n_u^2 - 1}{n_u^2}, \tag{1}$$

which uses the empirical value of α (through Δr_{val}) and thus ultimately the empirical value of G. A fully parameter-free derivation of G requires expressing ε_G in terms of the integers of the proton quintuplet without any empirical gravitational input.

The present paper reports a conjecture towards this goal, emerging from systematic numerical exploration. The conjecture expresses ε_G as a closed-form expression over $\mathbb{Q}(\sqrt{5})$ with 17 ppm accuracy, and its proof is reduced to the derivation of the proton-to-electron mass ratio μ from the PDL axioms — itself a central open problem of the programme. Section 2 documents the

convergence series. Section 3 states the conjecture formally with structural interpretation of each factor. Section 4 reduces the proof to the mass ratio problem. Section 5 introduces the bare/dressed mass distinction and its physical interpretation. Section 6 places the conjecture in the context of experimental uncertainties. Section 7 formulates the open problems.

Notation. Throughout, $\varphi = (1+\sqrt{5})/2$ is the golden ratio; $\kappa = R_{\text{surf}}/R_{\text{tot}} = 310\varphi/11017 \in \mathbb{Q}(\sqrt{5})$; $\mathcal{C} = 575/576$; $\mu_{\text{PDL}} = R_{\text{tot}}/6 = 11017/6$; $\mu_{\text{exp}} = 1836.152673430$ (CODATA 2022 [17]); $\varepsilon_G = (Gm_p^2/\hbar c)^{1/18}$ with G from CODATA 2022.

2 The convergence series towards the conjecture

The CODATA 2022 value of ε_G is:

$$\varepsilon_G = \left(\frac{G_{\text{CODATA}}\, m_p^2}{\hbar c}\right)^{1/18} = 0.00751940. \tag{2}$$

Starting from the natural structural unit $\kappa/6$ — the surface engagement fraction per electron-budget unit — the following convergence series was obtained by systematic symbolic exploration over $\mathbb{Q}(\sqrt{5})$:

Table 1: Convergence series towards ε_G from structural expressions. Each row refines the previous by one structural correction. All expressions are elements of $\mathbb{Q}(\sqrt{5})$ or $\mathbb{Q}$. The CODATA 2022 value is $\varepsilon_G = 0.00751940$.

Expression	Value	Discrepancy (ppm)
$\kappa/6$	0.00758813	9,141
$(\kappa/6)\cdot(1-\kappa/6)$	0.00753055	1,483
$\kappa\cdot\mathcal{C}\,/\,(6+\kappa)$	0.00751791	198
$\kappa\cdot\mathcal{C}\,/\,(6+\kappa)\cdot(1+2/R_{\text{tot}})$	0.00751927	**17**

Each step introduces a correction that is structurally motivated:

— $\kappa/6 \;\to\; (\kappa/6)(1-\kappa/6)$: first-order self-engagement correction, reducing the leakage by the fraction of the surface already engaged by the proton itself.

— $(\kappa/6)(1-\kappa/6) \;\to\; \kappa\mathcal{C}/(6+\kappa)$: replacement of the additive self-correction by the multiplicative coherence factor $\mathcal{C} = 575/576$, the same no-self-loop correction present in the bridge formula (1), combined with the renormalisation $6 \to 6+\kappa$ of the electron budget by the proton surface back-reaction. Reduction from 1,483 ppm to 198 ppm.

— $\kappa\mathcal{C}/(6+\kappa) \;\to\; \kappa\mathcal{C}/(6+\kappa)\cdot(1+2/R_{\text{tot}})$: second-order correction with factor $2/R_{\text{tot}}$. Reduction from 198 ppm to 17 ppm.

The factor of 2 in $2/R_{\text{tot}}$ is discussed in Section 3.

3 The conjecture

Conjecture 3.1 (Structural expression for ε_G)**.** The primary gravitational leakage parameter ε_G satisfies the exact identity over $\mathbb{Q}(\sqrt{5})$:

$$\varepsilon_G \;=\; \frac{\kappa\cdot\mathcal{C}}{6+\kappa}\cdot\left(1+\frac{2}{R_{\text{tot}}}\right), \tag{3}$$

where $\kappa = 310\varphi/11017 \in \mathbb{Q}(\sqrt{5})$, $\mathcal{C} = 575/576 \in \mathbb{Q}$, and $R_{\text{tot}} = 11017 \in \mathbb{Z}$. All quantities on the right-hand side are determined by the proton integer quintuplet $(24, 28, 930, 10087, 11017)$ without any empirical gravitational input.

3.1 Numerical evaluation

The right-hand side of (3) evaluates as follows:

$$\frac{\kappa \cdot \mathcal{C}}{6+\kappa} = \frac{(310\varphi/11017) \times (575/576)}{6 + 310\varphi/11017} = 0.00751791, \tag{4}$$

$$1 + \frac{2}{R_{\text{tot}}} = 1 + \frac{2}{11017} = 1.00018161, \tag{5}$$

$$\text{Product} = 0.00751791 \times 1.00018161 = 0.00751927. \tag{6}$$

Source	Value	Discrepancy
Conjecture (3) (this work)	0.00751927	17 ppm
ε_G from bridge formula (D24 [4])	0.00751942	2.4 ppm
ε_G CODATA 2022	0.00751940	—

The conjecture achieves 17 ppm accuracy. For comparison, the bridge formula calibrated from α_{exp} and G_{exp} gives 2.4 ppm. The residual gap of 17 ppm is structural in origin, as analysed in Section 4.

3.2 Structural interpretation of each factor

Each factor in equation (3) has a precise structural interpretation within the PDL relational hierarchy.

Factor κ: The surface engagement fraction $\kappa = R_{\text{surf}}/R_{\text{tot}} = 310\varphi/11017$ is the probability that a randomly selected internal relation of the proton belongs to its active surface $R_{\text{surf}} = 310\varphi$. It is the fundamental leakage rate per structural level [6].

Factor $\mathcal{C} = (n_u^2 - 1)/n_u^2 = 575/576$: The no-self-loop coherence correction on the up-quark core. This factor already appears in the bridge formula (1) of D24 [4]. It accounts for the fact that the $n_u^2 = 576$ internal up-core constraints include one diagonal (self-loop) term that does not contribute to the coherence leakage, reducing the effective coupling by 1/576.

Denominator $(6 + \kappa)$: The integer 6 is the relational budget of the minimal PDL closure K_4 (the electron prototype). The term $6 + \kappa$ is the total effective budget of one electron-level engagement unit augmented by the back-reaction of the proton surface: the proton's active surface κ partially re-engages the electron closure through which it is leaking, renormalising the electron budget from 6 to $6 + \kappa$.

Factor $(1 + 2/R_{\text{tot}})$: The second-order correction. The factor 2 reflects the two structurally distinct quark-core types in the proton — up-core ($n_u = 24$) and down-core ($n_d = 28$) — each contributing an independent secondary leakage channel of amplitude $1/R_{\text{tot}}$ through the total relational budget. The combined correction is $(1 + 1/R_{\text{tot}} + 1/R_{\text{tot}}) = (1 + 2/R_{\text{tot}})$. This factor contributes a reduction from 198 ppm to 17 ppm in the discrepancy with CODATA 2022.

4 Reduction of the proof to the mass ratio problem

4.1 The equivalence

We now show that Conjecture 3.1 is algebraically equivalent to the assertion that the physical proton-to-electron mass ratio takes a specific value $\mu^* \in \mathbb{Q}(\sqrt{5})$.

The bridge formula of D24 [4] gives:

$$\varepsilon_G^{\text{bridge}}(\mu) \;=\; \frac{\Delta r_{\text{val}}(\mu)}{\mathcal{R}\cdot\mathcal{C}}, \qquad \Delta r_{\text{val}}(\mu) = \frac{3}{\varphi}\sqrt{\frac{\mu}{\alpha_{\text{exp}}}} - r_{\text{val}}, \tag{7}$$

where α_{exp} is the experimental fine-structure constant. Setting $\varepsilon_G^{\text{bridge}}(\mu^*) = \varepsilon_G^B$ (the right-hand side of (3)) and solving for μ^* yields a unique positive solution:

Proposition 4.1. *The equation $\varepsilon_G^{\text{bridge}}(\mu^*) = \kappa\mathcal{C}/(6+\kappa)\cdot(1+2/R_{\text{tot}})$ has the unique positive solution*

$$\mu^* \;=\; 1836.152670 \;\in\; \mathbb{Q}(\sqrt{5}), \tag{8}$$

which satisfies:

$$|\mu^* - \mu_{\text{exp}}|/\mu_{\text{exp}} = 0.002 \text{ ppm}, \tag{9}$$

$$|\mu^* - \mu_{\text{PDL}}|/\mu_{\text{PDL}} = 7.62 \text{ ppm}, \tag{10}$$

where $\mu_{\text{exp}} = 1836.152673$ (CODATA 2022) and $\mu_{\text{PDL}} = R_{\text{tot}}/6 = 11017/6 = 1836.1\overline{6}$.

Corollary 4.2. *Conjecture 3.1 is true if and only if the physical mass ratio satisfies $\mu = \mu^*$, i.e. if and only if μ^* is the value of μ predicted by the* PDL *axioms.*

4.2 The three values of μ and their interpretation

Three distinct values of the mass ratio appear in the PDL analysis, and their relationships clarify the proof strategy:

Value	Expression	Status
$\mu_{\text{PDL}} = 11017/6$	$R_{\text{tot}}(p)/R_{\text{tot}}(e)$	Bare structural ratio; exact in $\mathbb{Z}$
$\mu^* = 1836.152670$	Solution of Conjecture 3.1	Required value; exact in $\mathbb{Q}(\sqrt{5})$
$\mu_{\text{exp}} = 1836.152673$	CODATA 2022 measurement	Dressed physical ratio

The three values are ordered: $\mu_{\text{PDL}} > \mu_{\text{exp}} \approx \mu^*$. The gap $\mu_{\text{PDL}} - \mu_{\text{exp}} = 0.01399$ (7.62 ppm) is the *structural dressing correction* discussed in Section 5. The gap $\mu_{\text{exp}} - \mu^* = 3.6\times10^{-6}$ (0.002 ppm) is the *residual gap* of the conjecture — four orders of magnitude below the current experimental precision on μ.

4.3 Proof strategy

The proof of Conjecture 3.1 decomposes into two steps:

Step 1 (algebraic): Show that $\mu^* \in \mathbb{Q}(\sqrt{5})$ is the unique value predicted by the PDL surface dressing mechanism. This requires deriving the correction $\delta\mu = \mu_{\text{PDL}} - \mu_{\text{phys}}$ from the proton–electron engagement geometry, i.e. from κ, $\kappa_e = \varphi/3$ (the surface engagement fraction of K_4), and the structural integers.

Step 2 (physical): Verify that μ^* is the value of μ extracted by the PDL axioms under the full dressing procedure, i.e. that the PDL prediction of μ is μ^* rather than μ_{PDL}.

If both steps are established, Conjecture 3.1 is proved, and ε_G — and hence G — becomes a fully combinatorial prediction of the proton quintuplet with no empirical gravitational input.

5 Bare and dressed mass ratio: structural interpretation of $\delta\mu$

5.1 The bare structural ratio μ_{PDL}

Within the PDL framework, the mass of a stable closure is its total relational budget. The minimal closure K_4 has $R_{\mathrm{tot}}(e) = 6$ (the six edges of the complete graph on four vertices). The proton has $R_{\mathrm{tot}}(p) = 11017$. The *bare* mass ratio is therefore:

$$\mu_{\mathrm{PDL}} \;=\; \frac{R_{\mathrm{tot}}(p)}{R_{\mathrm{tot}}(e)} \;=\; \frac{11017}{6} \;=\; 1836.1\overline{6}. \tag{11}$$

This is a purely combinatorial quantity, determined at zero coupling, without dynamic surface engagement.

5.2 The dressing correction $\delta\mu$

The experimentally measured mass ratio μ_{exp} is not the bare ratio but the *dressed* ratio, incorporating the dynamic coherence leakage between the proton and the electron through which it is observed. This dressing reduces μ_{PDL} to μ_{exp} by:

$$\delta\mu \;=\; \mu_{\mathrm{PDL}} - \mu_{\mathrm{exp}} \;=\; \frac{11017}{6} - 1836.15267 \;=\; 0.01399 \quad (7.62\ \mathrm{ppm}). \tag{12}$$

The dressing mechanism in PDL operates as follows. When a proton is observed via electromagnetic coupling to an electron closure, the proton's active surface $R_{\mathrm{surf}} = 310\varphi$ is partially engaged in this coupling. The fraction of the proton's total budget R_{tot} involved in the coupling is $\kappa = R_{\mathrm{surf}}/R_{\mathrm{tot}}$. The effective mass seen by the electron is therefore reduced:

$$\mu_{\mathrm{dressed}} \;\approx\; \mu_{\mathrm{PDL}} \cdot (1 - \kappa_{\mathrm{eff}}/6) \;=\; \mu_{\mathrm{PDL}} \cdot \left(1 - \frac{\kappa}{6+\kappa}\right), \tag{13}$$

where the factor 6 in the denominator reflects the electron budget $R_{\mathrm{tot}}(e) = 6$. The dressing correction thus involves the same combination $\kappa/(6+\kappa)$ that appears in Conjecture 3.1.

Remark 5.1. The measured value μ_{exp} includes all electromagnetic and QED corrections, and is obtained from spectroscopic measurements at finite temperature. A thermal correction to μ from the CMB background at $T = 2.725$ K would contribute approximately $\delta\mu/\mu \approx -k_B T/m_e c^2 \approx -7 \times 10^{-10}$, i.e. 0.0007 ppm — four orders of magnitude below the 7.62 ppm gap. The dressing correction $\delta\mu$ is therefore not of thermal origin but of structural (coherence-leakage) origin. At the epoch of recombination ($z \approx 1100$, $T \approx 3000$ K), the thermal correction would reach ≈ 0.75 ppm, still an order of magnitude below $\delta\mu$. The gap is a consequence of the PDL relational structure, not of the measurement environment.

5.3 Numerical test of the dressing model

The simple leading-order dressing model (13) gives:

$$\mu^{\mathrm{LO}}_{\mathrm{dressed}} \;=\; \frac{11017}{6} \cdot \left(1 - \frac{\kappa}{6+\kappa}\right) \;=\; 1836.1667 \times (1 - 0.00751791) \;=\; 1836.1529. \tag{14}$$

This is within 0.3 ppm of $\mu_{\mathrm{exp}} = 1836.1527$, confirming that the dressing model captures the correct order of magnitude and sign. A complete derivation would determine the exact dressed value from the PDL engagement geometry.

6 Experimental context

6.1 Experimental spread on G

Newton's gravitational constant G is the least precisely known fundamental constant. Recent measurements span a range of approximately 500 ppm, with individual quoted uncertainties of 11–150 ppm [14, 15, 12, 17]. A preliminary NIST measurement presented at the CPEM Conference in 2024 found a value below the CODATA recommended value with a combined uncertainty of 24.1 ppm [3], and the IUPAP Working Group on G has organised dedicated coordination meetings to address these discrepancies [16].

The PDL prediction $G_{\mathrm{PDL}} = 6.67448 \times 10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}}$ (D24 [4]) lies within the full spread of all modern measurements. The present conjecture, which gives ε_G at 17 ppm from CODATA, translates to an uncertainty on G of $18 \times 17 = 306$ ppm (the factor 18 arising from the power law $G \propto \varepsilon_G^{18}$). This 306 ppm discrepancy is well within the 500 ppm experimental spread. The conjecture is therefore fully compatible with the current experimental state of G measurements.

Table 2: Comparison of the PDL conjecture for G against independent experimental determinations. The conjecture gives $\varepsilon_G^B = 0.00751927$, hence $G^B = (\varepsilon_G^B)^{18}\,\hbar c/m_p^2$. The experimental spread of approximately 500 ppm encompasses the conjecture.

Source	G (10^{-11} $\mathrm{m^3\,kg^{-1}\,s^{-2}}$)	Deviation from G^B
PDL conjecture (this work)	6.6724	—
CODATA 2022 [17]	6.67430 ± 22 ppm	+284 ppm
Quinn et al. [14]	6.67545 ± 21 ppm	+457 ppm
Rosi et al. [15]	6.67191 ± 150 ppm	+68 ppm
Li et al. [12]	6.67484 & 6.67432	+373 & + 295 ppm
Full experimental spread	—	≈ 500 ppm

6.2 Experimental precision on μ

In contrast to G, the proton-to-electron mass ratio μ is one of the most precisely measured fundamental constants. The 2025 laser spectroscopy measurement on H_2^+ yielded $\mu = 1836.152673414(47)$ with a relative uncertainty of 2.6×10^{-11} [1, 13], i.e. 0.000026 ppm. The CODATA 2022 recommended value is $\mu = 1836.15267343(32)$ with a relative uncertainty of 1.7×10^{-11} [17].

The required value $\mu^* = 1836.152670$ differs from μ_{exp} by 3.6×10^{-6}, i.e. 0.002 ppm. This is approximately 100 times larger than the current measurement uncertainty on μ (0.000017 ppm). The conjecture therefore makes a falsifiable prediction: the PDL value of μ is μ^*, which differs from the current CODATA value at the level of 0.002 ppm. This discrepancy is outside the measurement uncertainty of μ and constitutes a genuine quantitative prediction that distinguishes the conjecture from a tautology.

Remark 6.1. The value $\mu^* = 1836.152670$ could in principle be tested directly against future ultra-high-precision measurements of μ at the sub-ppb level, independently of any measurement of G. This provides a route to testing the PDL conjecture that does not require gravitational experiments.

6.3 Cosmological constraints on G

An independent constraint on G from cosmological observables (Planck CMB, DESI BAO, and BBN) was recently reported at 1.8% precision, compatible with laboratory measurements at 1σ [2]. The PDL prediction $G_{\mathrm{PDL}} = 6.67448 \times 10^{-11}$ is also compatible with these cosmological

constraints. Within the PDL framework, the cosmological value of G is moreover expected to differ from the local value by up to 15%, as analysed in D26 [6] and D27 [9], with the local saturated value G_{PDL} corresponding to the dense galactic environment probed by laboratory experiments.

7 Epistemic status and open problems

7.1 Epistemic classification

Table 3: Epistemic classification of the results of this paper.

Result	Status	Basis
$\varepsilon_G^B = \kappa\mathcal{C}/(6+\kappa)\cdot(1+2/R_{\mathrm{tot}}) \in \mathbb{Q}(\sqrt{5})$	Exact (by construction)	This work
Numerical agreement with CODATA 2022 at 17 ppm	Verified	CODATA 2022 [17]
17 ppm $\ll$ 500 ppm experimental spread on G	Verified	[14, 15, 12, 3]
Conjecture iff $\mu = \mu^*$ (Proposition 4.1)	Proved algebraically	This work
μ^* agrees with μ_{exp} at 0.002 ppm	Verified	CODATA 2022 [17]
Leading-order dressing model $\mu^{\mathrm{LO}}_{\mathrm{dressed}}$ at 0.3 ppm	Numerical agreement	This work
Proof of Conjecture 3.1	**Open**	**OP1 below**

7.2 Open problems

OP1 (Proof of the conjecture). Prove equation (3) as an exact identity, either by deriving $\delta\mu = \mu_{\mathrm{PDL}} - \mu_{\mathrm{phys}}$ from the PDL proton–electron engagement geometry (algebraic route), or by computing the minimal frustration density $\eta(G_{p^*})$ of the proton constraint graph explicitly (graph-theoretic route). Either route would establish ε_G as a fully combinatorial prediction of the proton quintuplet, making G parameter-free.

OP2 (Structural origin of the factor $2/R_{\mathrm{tot}}$). Prove that the correction $(1+2/R_{\mathrm{tot}})$ arises from the two independent quark-core leakage channels, i.e. that the up-core ($n_u = 24$) and down-core ($n_d = 28$) each contribute $1/R_{\mathrm{tot}}$ to the second-order leakage amplitude independently.

OP3 (Derivation of $\delta\mu$ on $\mathbb{Q}(\sqrt{5})$). Derive the dressing correction $\delta\mu = 11017/6 - \mu_{\mathrm{phys}}$ as a closed-form expression over $\mathbb{Q}(\sqrt{5})$ from the PDL engagement geometry, using κ, $\kappa_e = \varphi/3$, and the structural integers. Verify that the leading-order model $\delta\mu \approx \mu_{\mathrm{PDL}}\cdot\kappa/(6+\kappa)$ can be extended to reproduce $\delta\mu$ to the full precision of μ measurements ($\sim 10^{-11}$).

OP4 (Observational test via μ measurements). The conjecture predicts $\mu = \mu^* = 1836.152670$, differing from $\mu_{\mathrm{exp}} = 1836.152673$ at 0.002 ppm. This discrepancy is approximately 100 times larger than the current measurement uncertainty on μ. A dedicated comparison of μ^* against ultra-precision spectroscopic determinations of μ would provide an independent test of the conjecture that does not require any measurement of G.

OP5 (Higher-order corrections to the bridge formula). The residual 0.34% in the bridge formula (D24 [4]) and the 17 ppm gap in the present conjecture may share a common structural origin. Identifying whether the series $\kappa/6 \to \kappa\mathcal{C}/(6+\kappa) \to \kappa\mathcal{C}/(6+\kappa)\cdot(1+2/R_{\mathrm{tot}})$ terminates at

the next term, and whether the next term reduces the residual below 1 ppm, would substantially strengthen Conjecture 3.1.

Conclusion

This paper has presented a structural conjecture for the primary gravitational leakage parameter ε_G of the PDL framework. The conjecture (3) expresses ε_G entirely in terms of structural quantities of the proton quintuplet — $\kappa = 310\varphi/11017$, $\mathcal{C} = 575/576$, and $R_{\text{tot}} = 11017$ — without any empirical gravitational input, and achieves 17 ppm accuracy against CODATA 2022. This residual is within the 500 ppm experimental spread of independent measurements of G.

The convergence series (Table 1) documents the structural refinement from the natural unit $\kappa/6$ to the full conjecture in four steps, each with a clear physical interpretation. The proof is reduced to a single open problem: the derivation of the proton-to-electron mass ratio μ from the PDL axioms. The required value $\mu^* = 1836.152670$ agrees with the CODATA measurement to 0.002 ppm, providing a testable quantitative prediction.

The bare/dressed mass distinction clarifies the structure of the problem: $\mu_{\text{PDL}} = 11017/6$ is the structurally exact bare ratio; μ_{exp} is the dressed ratio including coherence-leakage corrections; and the dressing correction $\delta\mu = 0.01399$ is the surface engagement contribution of $\kappa/(6+\kappa)$ applied to the proton–electron coupling. The leading-order model reproduces μ_{exp} to 0.3 ppm.

If Conjecture 3.1 is proved, ε_G becomes a purely combinatorial quantity, the bridge formula of D24 predicts G from the proton quintuplet alone, and the fundamental coupling constants α, G, and k_B all emerge from the single combinatorial object $(24, 28, 930, 10087, 11017)$ without any empirical input from the gravitational sector.

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